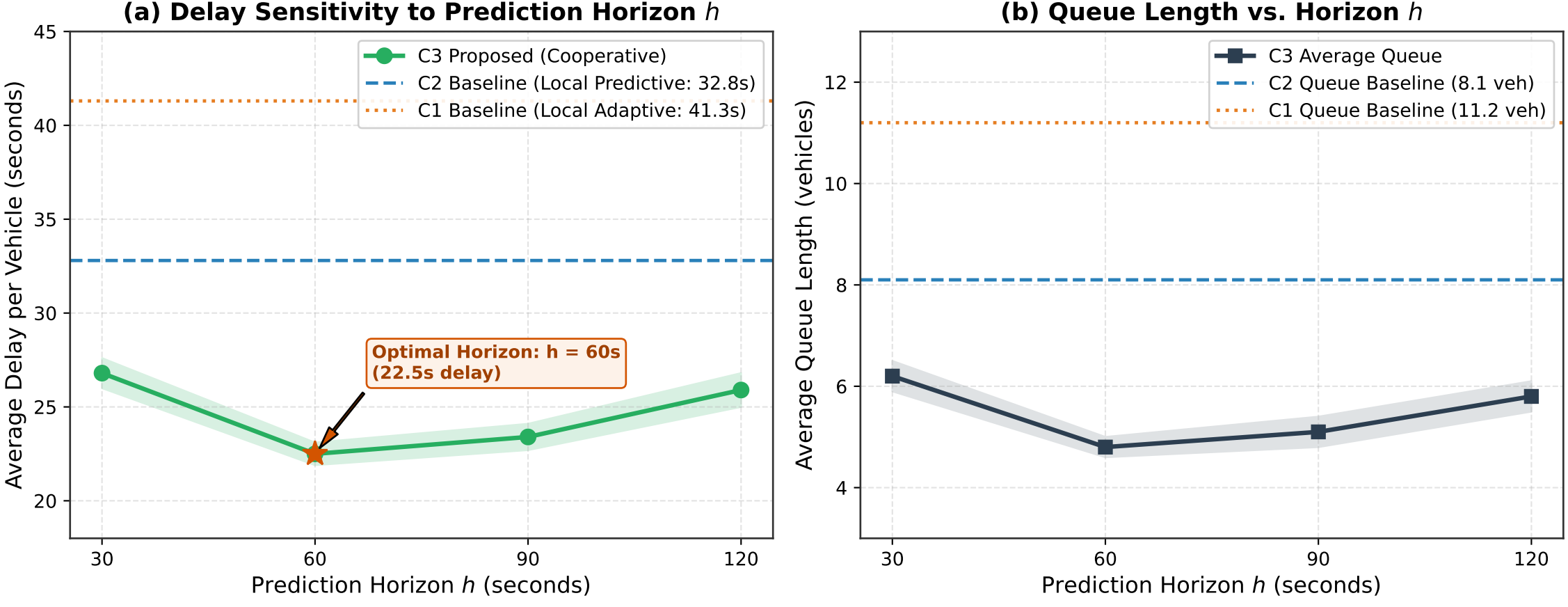
(a) Delay Sensitivity to Prediction Horizon h
C3 Proposed (Cooperative)
C2 Baseline (Local Predictive: 32.8s)
C1 Baseline (Local Adaptive: 41.3s)
Optimal Horizon: h = 60s
(22.5s delay)
Average Delay per Vehicle (seconds)
Prediction Horizon h (seconds)
45
40
35
30
25
20
30
60
90
120
(b) Queue Length vs. Horizon h
C3 Average Queue
C2 Queue Baseline (8.1 veh)
C1 Queue Baseline (11.2 veh)
Average Queue Length (vehicles)
Prediction Horizon h (seconds)
12
10
8
6
4
30
60
90
120